

AZIM PREMJI UNIVERSITY, BHOPAL

First Semester 2026–2027

Tutorial Sheet–8

Course No. MTH-113

Course title: Introduction to Quantitative Thinking

- (1) In a Mathematics test, the following marks were obtained by 40 students. Arrange these marks in a table using tally marks.

8	1	3	7	6	5	5	4	4	2
4	9	5	3	7	1	6	5	2	7
7	3	8	4	2	8	9	5	8	6
7	4	5	6	9	6	4	4	6	6

- (a) Find how many students obtained marks equal to or more than 7.
(b) How many students obtained marks below 4?
- (2) Following is the choice of sweets of 30 students of Class VI.
Ladoo, Barfi, Ladoo, Jalebi, Ladoo, Rasgulla, Jalebi, Ladoo, Barfi, Rasgulla, Ladoo, Jalebi, Jalebi, Rasgulla, Ladoo, Rasgulla, Jalebi, Ladoo, Rasgulla, Ladoo, Ladoo, Barfi, Rasgulla, Rasgulla, Jalebi, Rasgulla, Ladoo, Rasgulla, Jalebi, Ladoo.
- (a) Arrange the names of sweets in a table using tally marks.
(b) Which sweet is preferred by most of the students?
- (3) Catherine threw a dice 40 times and noted the number appearing each time as shown below:

1	3	5	6	6	3	5	4	1	6
2	5	3	4	6	1	5	5	6	1
1	2	2	3	5	2	4	5	5	6
5	1	6	2	3	5	2	4	1	5

Make a table and enter the data using tally marks.

- (a) Find the number that appeared the minimum number of times
(b) Find the number that appeared the maximum number of times
(c) Find those numbers that appear an equal number of times.
- (4) The number of girl students in each class of a co-educational middle school is depicted by the pictograph:

Classes	Number of girl students	 - 4 Girls
I	     	
II	     	
III	     	
IV	    	
V	  	
VI	   	
VII	  	
VIII	 	

Observe this pictograph and answer the following questions:

- (a) Which class has the minimum number of girl students?
 - (b) Is the number of girls in Class VI less than the number of girls in Class V?
 - (c) How many girls are there in Class VII?
- (5) A survey of 120 school students was done to find which activity they prefer to do in their free time.

Preferred activity	Number of students
Playing	45
Reading story books	30
Watching TV	20
Listening to music	10
Painting	15

Draw a bar graph to illustrate the above data taking scale of 1 unit length = 5 students. Which activity is preferred by most of the students other than playing?

- (6) The number of Mathematics books sold by a shopkeeper on six consecutive days is shown below:

Days	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday
Number of Books Sold	65	40	30	50	20	70

Draw a bar graph to represent the above information choosing the scale of your choice.

- (7) Following table shows the number of bicycles manufactured in a factory during the years 1998 to 2002. Illustrate this data using a bar graph. Choose a scale of your choice.

Years	Number of bicycles manufactured
1998	800
1999	600
2000	900
2001	1100
2002	1200

- (a) In which year were the maximum number of bicycles manufactured?
 - (b) In which year were the minimum number of bicycles manufactured?
- (8) Number of persons in various age groups in a town is given in the following table.

Age group	1-14	15-29	30-44	45-59	60-74	75 and above
Number of Persons	2 lakhs	1 lakhs 60 thousands	1 lakhs 20 thousands	1 lakhs 20 thousands	80	40 thousands

Draw a bar graph to represent the above information and answer the following questions. (take 1 unit length = 20 thousands)

- (a) Which two age groups have same population?
 - (b) All persons in the age group of 60 and above are called senior citizens. How many senior citizens are there in the town?
- (9) For which of these would you use a histogram to show the data?
- (a) The number of letters for different areas in a postman's bag.
 - (b) The height of competitors in an athletics meet.
 - (c) The number of cassettes produced by 5 companies.
 - (d) The number of passengers boarding trains from 7:00 a.m. to 7:00 p.m. at a station.

Give reasons for each.

- (10) The weekly wages (in Rs) of 30 workers in a factory are.
830, 835, 890, 810, 835, 836, 869, 845, 898, 890, 820, 860, 832, 833, 855, 845, 804, 808, 812, 840, 885, 835, 835, 836, 878, 840, 868, 890, 806, 840.

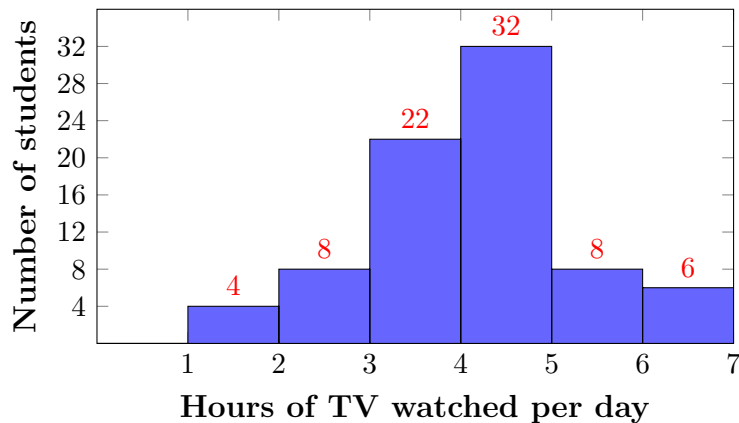
Draw a histogram for the frequency table made for the above data, and answer the following questions.

- Which group has the maximum number of workers?
- How many workers earn Rs 850 and more?
- How many workers earn less than Rs 850?

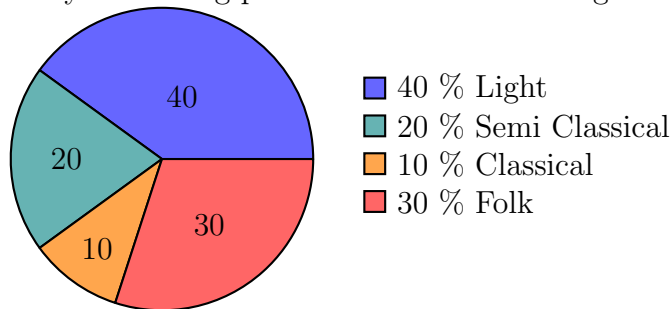
- (11) The number of hours for which students of a particular class watched television during holidays is shown through the given graph.

Answer the following.

- For how many hours did the maximum number of students watch TV?
- How many students watched TV for less than 4 hours?
- How many students spent more than 5 hours in watching TV?



- (12) A survey was made to find the type of music that a certain group of young people liked in a city. Following pie chart shows the findings of this survey.



From this pie chart answer the following:

- If 20 people liked classical music, how many young people were surveyed?
 - Which type of music is liked by the maximum number of people?
 - If a cassette company were to make 1000 CD's, how many of each type would they make?
- (13) Draw a pie chart showing the following information. The table shows the colours preferred by a group of people.

Colours	Number of People
Blue	18
Green	9
Red	6
Yellow	3
Total	36

(14) List the outcomes you can see in these experiments.

(a) Spinning a wheel

(b) Tossing two coins together



(15) When a die is thrown, list the outcomes of an event of getting

(a) a prime number

(b) not a prime number

(c) a number greater than 5

(d) a number not greater than 5.

(16) Find the.

(a) Probability of the pointer stopping on D in (Question 14-(a))?

(b) Probability of getting an ace from a well shuffled deck of 52 playing cards?

(c) Probability of getting a red apple. (See figure below)

